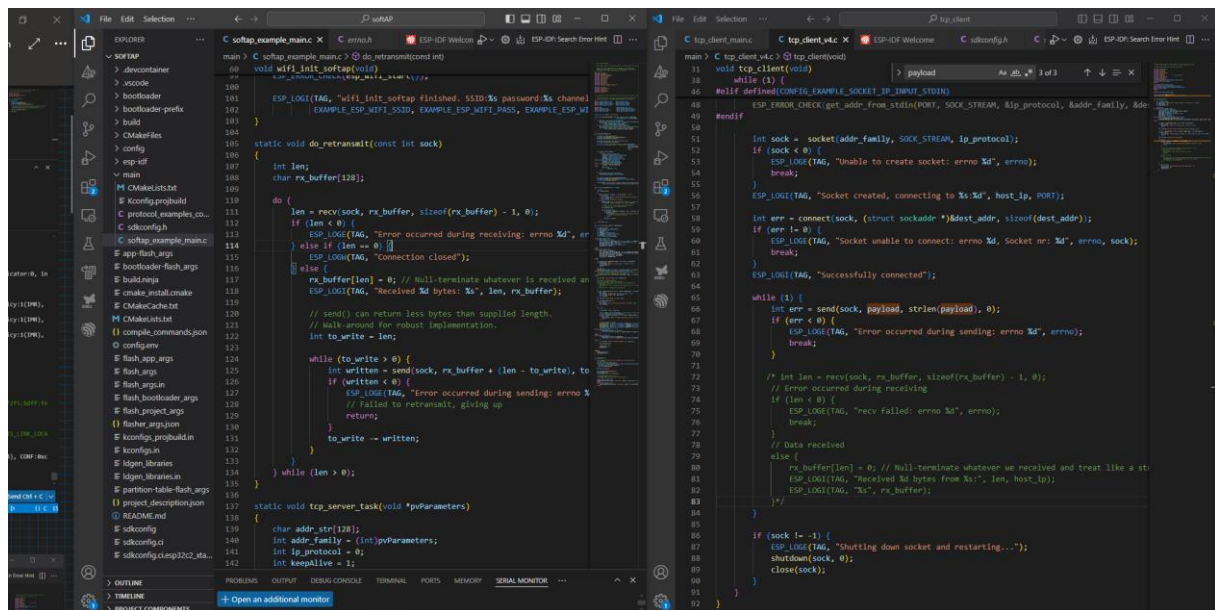
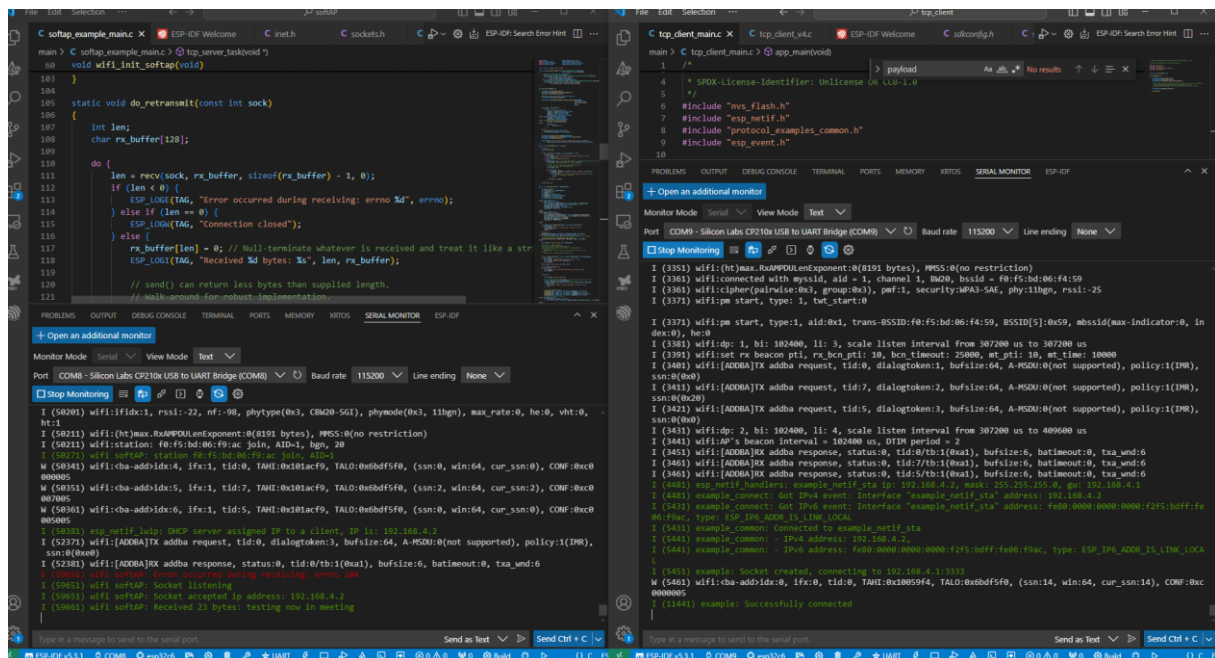


How the code of AP (TCP host) and Sta (TCP client) is transmitting/receiving data :

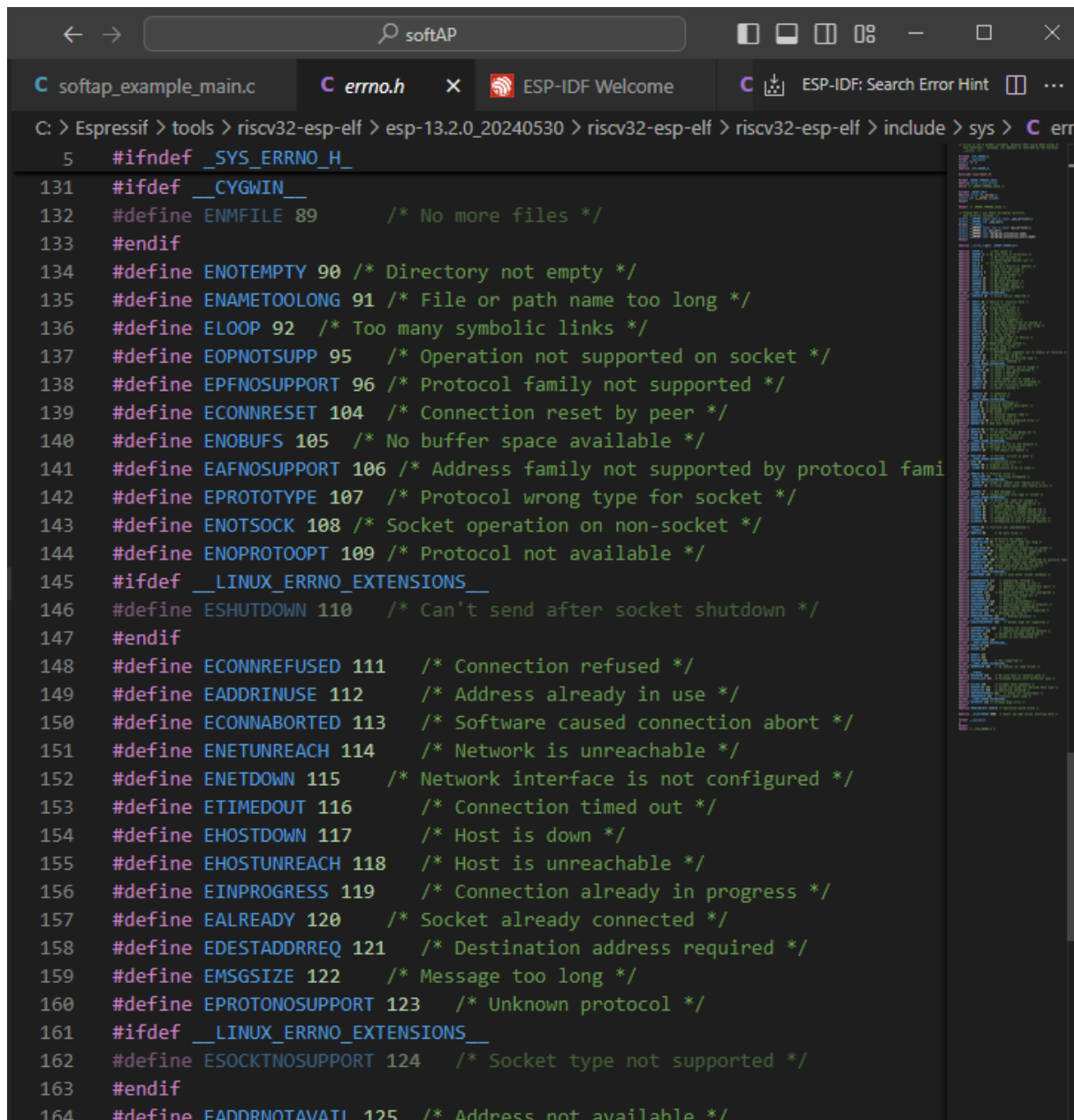


- STA (right) sends payload with send() and awaits retransmission with recv()
- AP (left) receives payload, transmits it back to STA (right)

Different scenarios when I disable certain functions :



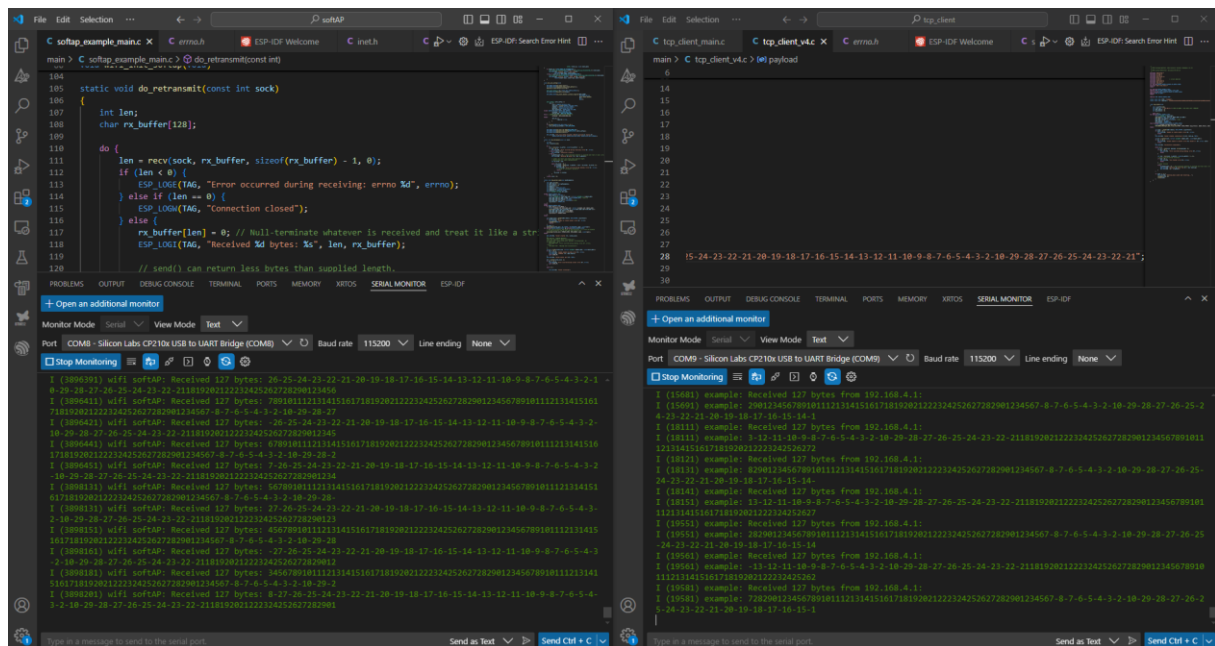
- send() of AP is disabled; we receive the payload at AP only once (I assume TCP-client on the left is waiting for a transmission that never comes)



```
5  #ifndef _SYS_ERRNO_H_
131 #ifdef __CYGWIN__
132 #define ENMFILE 89 /* No more files */
133 #endif
134 #define ENOTEMPTY 90 /* Directory not empty */
135 #define ENAMETOOLONG 91 /* File or path name too long */
136 #define ELOOP 92 /* Too many symbolic links */
137 #define EOPNOTSUPP 95 /* Operation not supported on socket */
138 #define EPNOSUPPORT 96 /* Protocol family not supported */
139 #define ECONNRESET 104 /* Connection reset by peer */
140 #define ENOBUFS 105 /* No buffer space available */
141 #define EAFNOSUPPORT 106 /* Address family not supported by protocol fami
142 #define EPROTOTYPE 107 /* Protocol wrong type for socket */
143 #define ENOTSOCK 108 /* Socket operation on non-socket */
144 #define ENOPROTOPT 109 /* Protocol not available */
145 #ifdef __LINUX_ERRNO_EXTENSIONS__
146 #define ESHUTDOWN 110 /* Can't send after socket shutdown */
147 #endif
148 #define ECONNREFUSED 111 /* Connection refused */
149 #define EADDRINUSE 112 /* Address already in use */
150 #define ECONNABORTED 113 /* Software caused connection abort */
151 #define ENETUNREACH 114 /* Network is unreachable */
152 #define ENETDOWN 115 /* Network interface is not configured */
153 #define ETIMEDOUT 116 /* Connection timed out */
154 #define EHOSTDOWN 117 /* Host is down */
155 #define EHOSTUNREACH 118 /* Host is unreachable */
156 #define EINPROGRESS 119 /* Connection already in progress */
157 #define EALREADY 120 /* Socket already connected */
158 #define EDESTADDRREQ 121 /* Destination address required */
159 #define EMSGSIZE 122 /* Message too long */
160 #define EPROTONOSUPPORT 123 /* Unknown protocol */
161 #ifdef __LINUX_ERRNO_EXTENSIONS__
162 #define ESOCKTNOSUPPORT 124 /* Socket type not supported */
163 #endif
164 #define EADDRNOTAVAIL 125 /* Address not available */
```

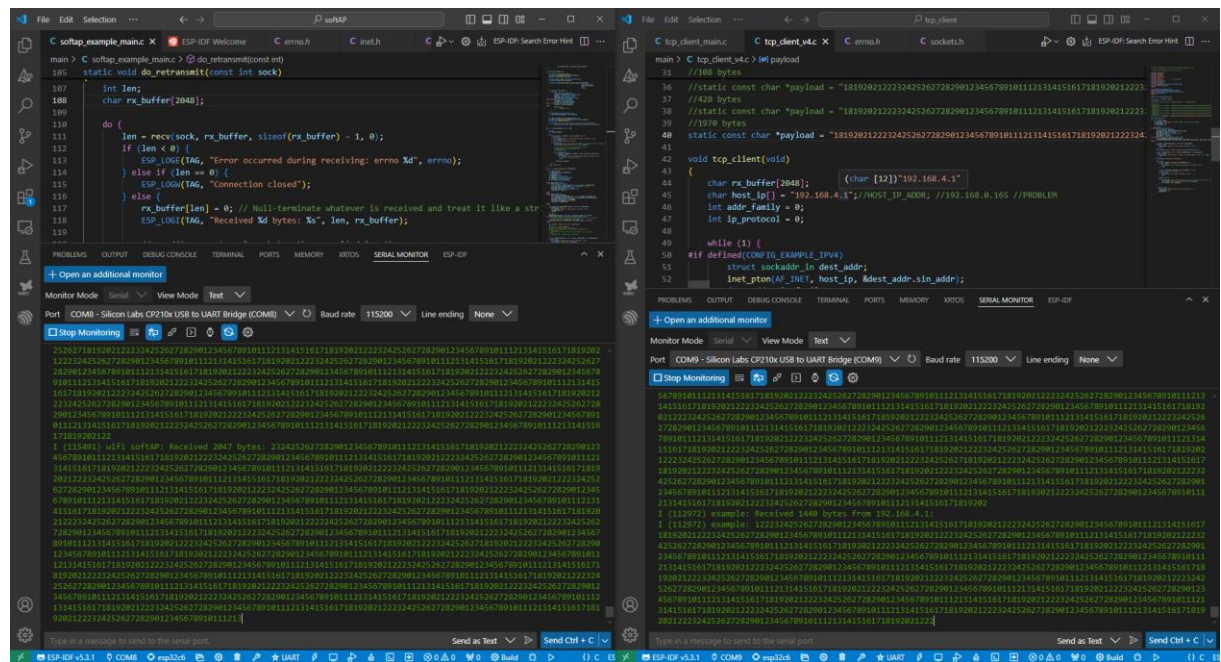
Defined errors

I also ran iPerf again and found that about 345'358 bytes get transmitted before the error occurs



I've increased the payload's size to 127 bytes (just like in iPerf). Here it also stops at some point. The state you see in the pic has been now for a while (with the previous smaller payload it never stalled)

So payload size could also be an issue



- I wanted to push it to the limit so I created a string of size 1970 bytes
- Transmission does work, continuously even (in the image it gets fragmented which wasn't the case the first time I tried)
- There are errors though (stack overflow; see pics below); but transmissions get picked up again; therefore short breaks but communication still flows afterwards

The screenshot shows an IDE with the following components:

- Editor:** Displays `softap_example_main.c` with the following code:


```

main > C softap_example_main.c > do_retransmit(const int)
60 void wifi_init_softap(void)
75     wifi_config_t wifi_config = {
76         .ap = {
82     #ifdef CONFIG_ESP_WIFI_SOFTAP_SAE_SUPPORT
84         .sae_pwe_h2e = WPA3_SAE_PWE_BOTH,
85     #else /* CONFIG_ESP_WIFI_SOFTAP_SAE_SUPPORT */
86         .authmode = WIFI_AUTH_WPA2_PSK,
87     #endif
88         .pmf_cfg = {
89             .required = true,
90         },
91     },
92 };
93 if (strlen(EXAMPLE_ESP_WIFI_PASS) == 0) {
94     wifi_config.ap.authmode = WIFI_AUTH_OPEN;
      
```
- Serial Monitor:**
 - Port: COM8 - Silicon Labs CP210x USB to UART Bridge (COM8)
 - Baud rate: 115200
 - Line ending: None
 - Start Monitoring
- Output:**

```

72829012345678910111213141516171819202122232425262728290123456789101112131415161718192021222324252627282901234567
89101112131415161718192021222324252627282901234567891011121314151617181920212223242526272829012345678910111213141
516171819202122232425262728290123
I (58540) wifi softAP: Received 205012 bytes (in total): 18192021222324252627282901234567891011121314151617181920
21222324252627282901234567891011121314151617181920212223242526272829012345678910111213141516171819202122232425262
72829012345678910111213141516171819202122232425262728290123456789101112131415161718192021222324252627282901234567
89101112131415161718192021222324252627282901234567891011121314151617181920212223242526272829012345678910111213141
516171819202122232425262728290123
***ERROR*** A stack overflow in task tcp_server has been detected.
Core 0 register dump:
MEPC   : 0x408008a6  RA    : 0x408075d2  SP    : 0x4081cbf0  GP    : 0x40818f44
TP     : 0x40835ac0  T0    : 0x40030dca  T1    : 0x0000000f  T2    : 0x00000002
S0/FP  : 0x00000003  S1    : 0x80000005  A0    : 0x4081cc08  A1    : 0x4209a824
A2     : 0x00000003  A3    : 0x4081cc38  A4    : 0x00000001  A5    : 0x40821000
A6     : 0x420092ee  A7    : 0x0000000a  S2    : 0x00001881  S3    : 0x00000001
S4     : 0x00000000  S5    : 0x40834db0  S6    : 0x40835180  S7    : 0x408213bc
S8     : 0x000001ac  S9    : 0x00000000  S10   : 0x4080028a  S11   : 0x00001881
T3     : 0x00000000  T4    : 0x00000000  T5    : 0x00000000  T6    : 0x00000000
MSTATUS: 0x00001801  MTVEC  : 0x40800001  MCAUSE : 0x00000007  MTVAL  : 0x00000000
MHARTID : 0x00000000

Stack memory:
4081cbf0: 0x00000000 0x00000000 0x00000000 0x40808946 0x00000000 0x00000001 0x452a2a2a 0x524f5252
4081cc10: 0x202a2a2a 0x74732041 0x206b6361 0x7265766f 0x776f6c66 0x206e6920 0x6b736174 0x70637420
4081cc30: 0x7265735f 0x20726576 0x20736168 0x6e656562 0x74656420 0x65746365 0x00002e64 0x00000000
      
```

- On AP/Host side

I also tried measuring the throughput 'by hand' by letting the AP count the received bytes in total and measuring the time with my phone. Highest average throughput was about 10KB/s. I'm not sure if it's representative, since the serial monitoring could slow the process down if the chip has to display each transmitted payload (display time is prob quite long).